How do we obtain the smallest system possible for dealing with control, observations, or representations?

I Minimality

A) McMillan Degree

Given the transfer function $G(s)$, the McMillan degree is the smallest value s.t. A is $n \times n$ and

$$G(s) = C(sI - A^{-1})B$$

$\delta(G)$

B) McMillan Realization

Given $\dot{x} = Ax + Bu$ $y = Cx$ IF $x \in \mathbb{R}^n$ with $n = \delta(G)$, then (A, B, C) is a minimal realization of G.

C) Theorem (Kalman)

(A, B, C) is minimal iff (A, B) is C.C. and (A, C) is C.O.

↳ can use observable design (later)
can use control design
} can control system

II Question of The Day

How do we make the system better in the sense of

$$G(s) = C(sI - A)^{-1}B = C'(sI - A')^{-1}B'$$

i) $(A, B, C) \xrightarrow{?} (A', B', C')$ C.C.
ii) $(A, B, C) \xrightarrow{?} (A', B', C')$ C.O.
} we have done these

iii) $(A, B, C) \xrightarrow{?} (A', B', C')$ C.C. & C.O.

A) (i) C.C.

$T = [\underbrace{v_1 \cdots v_d}_{\text{form basis for } Im(\mathcal{C})} \; \underbrace{u_1 \cdots u_{n-d}}_{\text{vectors that compliment } v_1 \cdots v_d}]$

We get the system

$y = \overset{\hat{C}}{CT}\hat{x} \quad \hat{C} = [\hat{C}_1 \; \hat{C}_2]$

$$\dot{x} = T^{-1}AT\hat{x} + T^{-1}Bu = \hat{A}\hat{x} + \hat{B}u = \begin{bmatrix} \hat{A}_{11} & A_{12} \\ 0 & \hat{A}_{22} \end{bmatrix} x + \begin{bmatrix} \hat{B}_1 \\ 0 \end{bmatrix} u$$

(A, B, C) is algebraically equivalent to $(\hat{A}, \hat{B}, \hat{C})$

(A, B, C) is zero-state equivalent to $(\hat{A}_{11}, \hat{B}_1, \hat{C}_1)$

Given an (A, B, C) that is not C.C, we can always find a ~~sys~~ sub-system that is. (However, we cannot do anything if the uncontrollable part is not stabilizeable)

B) (ii) C.O.

$v_1 \cdots v_d$ basis for $N(\mathcal{O})$
$u_1 \cdots u_{n-d}$ compliment $v_1 \cdots v_d$ } Basis $\mathbb{R}^n$

$T = u_1 \cdots u_{n-d} \; v_1 \cdots v_d]$

$\hat{A} = T^{-1}AT \quad \hat{C} = CT \quad B = T^{-1}B = \begin{bmatrix} \hat{B}_1 \\ \hat{B}_2 \end{bmatrix}$

$$\hat{A} = \begin{bmatrix} \hat{A}_{11} & 0 \\ \hat{A}_{21} & \hat{A}_{22} \end{bmatrix} \qquad \hat{C} = [\hat{C}_1 \quad 0]$$

(A, B, C) is algebraically equivalent to $(\hat{A}, \hat{B}, \hat{C})$

(A, B, C) is zero-state equivalent to $(\hat{A}_{11}, \hat{B}_1, \hat{C}_1)$ which is C.O.

Given (A, B, C) not C.O. we can always find a smaller C.O. that we can work with (However we have a similar caveat to (i)).

c) (iii) Minimality

$\dot{x} = Ax + Bu \qquad y = Cx$

C: Controllable

$\bar{C}$: Uncontrollable

O: Observable

$\bar{O}$: Unobservable

$$T = \left[\underbrace{[\,C\ O\,], [\,C\ \bar{O}\,]}_{\text{Controllable}} \,\vdots\, \underbrace{[\,\bar{C}\ O\,], [\,\bar{C}\ \bar{O}\,]}_{\text{Uncontrollable}} \right]$$

$$\mathbb{R}^n = \text{span}\{C\ O\} \oplus \text{span}\{C\ \bar{O}\} \oplus \text{span}\{\bar{C}\ O\} \oplus \text{span}\{\bar{C}\ \bar{O}\}$$

(This a controllable decomposition where we organize the columns Controllable | Uncontrollable)

Recall $\text{Im}(\Gamma)$: gives controllable subspace

$N(\Omega)$: gives unobservable subspace

$\text{span}\{C\ O\}$: $\text{Im}(\Gamma) \cap N(\Omega)^{\perp}$

$\text{span}\{C\ \bar{O}\}$: $\text{Im}(\Gamma) \cap N(\Omega)$

$\text{span}\{\bar{C}\ O\}$: $\text{Im}(\Gamma)^{\perp} \cap N(\Omega)$

$\text{span}\{\bar{C}\ \bar{O}\}$: $\text{Im}(\Gamma)^{\perp} \cap N(\Omega)^{\perp}$

1) Similiar to before, we are going to form a basis

2) Use this to form a new state

$$\hat{x} = T^{-1} x = \begin{bmatrix} \hat{x}_{co} \\ \hat{x}_{c\bar{o}} \\ \hat{x}_{\bar{c}o} \\ \hat{x}_{\bar{c}\bar{o}} \end{bmatrix} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \hat{x}_4 \end{bmatrix}$$

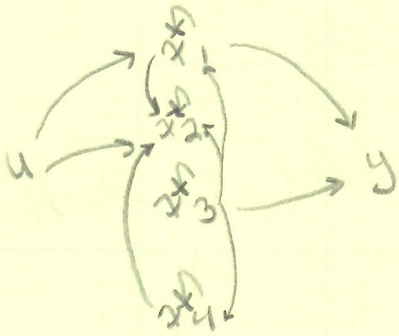
3) And dynamics

$$\dot{\hat{x}} = \hat{A}\hat{x} + \hat{B}u \qquad y = \hat{C}\hat{x}$$

$$\hat{A} = \begin{bmatrix} \hat{A}_{11} & 0 & \hat{A}_{13} & 0 \\ \hat{A}_{21} & \hat{A}_{22} & \hat{A}_{23} & \hat{A}_{24} \\ 0 & 0 & \hat{A}_{33} & 0 \\ 0 & 0 & \hat{A}_{43} & \hat{A}_{44} \end{bmatrix} \qquad \hat{B} = \begin{bmatrix} \hat{B}_1 \\ \hat{B}_2 \\ 0 \\ 0 \end{bmatrix} \qquad \hat{C} = \begin{bmatrix} \hat{C}_1 & 0 & \hat{C}_2 & 0 \end{bmatrix}$$

4) A Look at Interactions

Direct Affects



$\hat{x}_1$: (x_{co}) affected by x_1, x_3, u

$\hat{x}_2$: $(x_{c\bar{o}})$ affected by x_1, x_2, x_3, u

$\hat{x}_3$: $(x_{\bar{c}o})$ affected by x_3

$\hat{x}_4$: $(x_{\bar{c}\bar{o}})$ affected by x_3, x_4

y: affected by x_1, x_3

x_1 is indirectly affected by nothing

x_2 is indirectly affected by nothing

x_3 is indirectly affected by nothing

x_4 is indirectly affected by nothing

y is indirectly affected by u

only additional indirect affect added

There is no path between input and uncontrollable state

There is no path between unobservable states and the output.

D) Result and Example

1) Proposition

The realization $(\hat{A}_{11}, \hat{B}_1, \hat{C}_1)$ is zero-state equivalent to (A, B, C)

Moreover, it is a minimal representation ($\hat{A}_{11} \in \mathbb{R}^{\hat{n} \times \hat{n}}$ where $\hat{n} = \delta(G)$)

↳ Both C.C & C.O.

2) Example

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ -0.5 & 1 & -0.5 & 0 \\ 0 & -2 & -2 & -1 \\ -0.5 & 0 & -0.5 & -1 \end{bmatrix} \quad B = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \end{bmatrix} \quad C = \begin{bmatrix} 0.5 & -1 & 0.5 & 0 \end{bmatrix}$$

$$\Gamma = \begin{bmatrix} -1 & 6 & -8 & 20 \\ 2 & 2 & 2 & 2 \\ 1 & -6 & 8 & -20 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \Omega = \begin{bmatrix} 0.5 & -1 & 0.5 & 0 \\ 1 & -1 & 1 & 0 \\ 1.5 & -1 & 1.5 & 0 \\ 2 & -1 & 2 & 0 \end{bmatrix}$$

$\text{rank}(\Gamma) = 2$ $\qquad$ $\text{rank}(\Omega) = 2$

<u>Not C.C</u> $\qquad$ <u>Not C.O</u>

Therefore to decompose we need

$$\mathrm{Im}(\Gamma) \quad \underbrace{\mathrm{Im}(\Gamma)^{\perp}}_{N(\Gamma^T)} \quad N(\Omega) \quad \underbrace{N(\Omega)^{\perp}}_{\mathrm{Im}(\Omega^T)}$$

In MATLAB you can use

a) $\mathrm{Im}(\Gamma)$

orth(Γ) → MATLAB gives something not very nice, but is simplifies to

$$\mathrm{span}\{\mathrm{orth}(\Gamma)\} = \mathrm{span}\left\{ \underset{v_1}{\begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}} \underset{v_2}{\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}} \right\}$$

b) $\mathrm{Im}(\Gamma)^{\perp} = N(\Gamma^T)$

$$\mathrm{null}(\Gamma^T) = \begin{bmatrix} 0.7 & 0 \\ 0 & 0 \\ 0.7 & 0 \\ 0 & 1 \end{bmatrix} = \mathrm{span}\left\{ \overset{v_3}{\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}} \overset{v_4}{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}} \right\}$$

c) $N(\Omega)$

$$\mathrm{null}(\Omega) = \begin{bmatrix} 0.7 & 0 \\ 0 & 0 \\ -0.7 & 0 \\ 0 & 1 \end{bmatrix} = \mathrm{span}\left\{ \overset{v_1}{\begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}} \overset{v_4}{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}} \right\}$$

d) $N(\Omega)^{\perp} = \mathrm{Im}(\Omega^T)$

$$\mathrm{orth}(\Omega^T) = \begin{bmatrix} -0.6 & -0.3 \\ 0.4 & -0.9 \\ -0.6 & -0.3 \\ 0 & 0 \end{bmatrix} = \mathrm{span}\left\{ \overset{v_3}{\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}} \overset{v_2}{\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}} \right\}$$

$\angle O$ $\quad Im(\Gamma) \cap N(\Omega)^{\perp} = span\{v_2\}$

$\angle \bar{O}$ $\quad Im(\Gamma) \cap N(\Omega) = span\{v_1\}$

$\bar{\angle} O$ $\quad Im(\Gamma)^{\perp} \cap N(\Omega)^{\perp} = span\{v_3\}$

$\bar{\angle} \bar{O}$ $\quad Im(\Gamma)^{\perp} \cap N(\Omega) = span\{v_4\}$

e) Now form our transformed system

$$T = [v_2 \quad v_1 \quad v_3 \quad v_4]$$

$$= \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\hat{A} = T^{-1} A T = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 2 & -2 & 3 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & -1 \end{bmatrix}$$

$$\hat{B} = T^{-1} B = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 0 \end{bmatrix} \qquad \hat{C} = CT = [-1 \quad 0 \quad 1 \quad 0]$$

f) Extract minimal system

$$\hat{A}_{11} = [1] \qquad \hat{B}_1 = [2] \qquad \hat{C}_1 = [-1]$$

$$\hat{x}_1 = \hat{x}_1 + 2u \qquad y = -\hat{x}_1$$

zero-state equivalent w/ original system

↳ Is this a unique representation?

g) Take a look at Laplace transform

$$G(s) = \hat{C}_1 (sI - \hat{A}_{11}) \hat{B}_1 = -\frac{2}{s-1}$$

~ Is the Laplace transform unique?

All minimal realizations of a transfer function are algebraically equivalent